Shubham Farera



Senior Software Engineer (SDE-3) • Distributed Systems & Backend Architect • Java, Spring Boot & Kafka Specialist

Pune, India, 411021 +91 9584368070 shubham.farera@gmail.com LinkedIn GitHub

EXECUTIVE PROFESSIONAL SUMMARY

Senior Software Engineer with 8.5+ years of experience architecting, designing, and delivering high-throughput, fault-tolerant distributed systems and enterprise microservices. Deep expertise across High-Level Design (HLD) and Low-Level Design (LLD), applying SOLID principles, GoF design patterns, and clean architecture to scale production systems to 10M+ daily transactions with 99.99% uptime.

Proven specialist in distributed systems engineering: architecting event-driven pipelines with Apache Kafka, implementing distributed transactions via the Saga Pattern (Orchestration & Choreography) and Outbox Pattern, engineering low-latency search with Elasticsearch, and scaling databases via sharding and master-slave replication. Adept in navigating CAP and PACELC trade-offs to deliver resilient, sub-150ms latency platforms across Healthcare, FinTech, and Logistics.

TECHNICAL COMPETENCIES & ARCHITECTURE

- **</> Languages & Core :** **Java (8 / 11 / 17)**, Data Structures & Algorithms (DSA), Object-Oriented Analysis & Design (OOAD), Multithreading & Concurrency, Memory Management & JVM Tuning.
- **Distributed Systems :** **Apache Kafka** (Event Streaming, Partitioning), Saga Pattern (Orchestration & Choreography), Transactional Outbox, **Elasticsearch**, Redis Caching, Consistent Hashing, CAP & PACELC Theorems, ZooKeeper.
- **Architecture & LLD :** **Microservices Architecture**, High-Level Design (HLD), Low-Level Design (LLD), **SOLID Principles**, GoF Design Patterns (Strategy, Factory, Observer, Decorator, State), Domain-Driven Design (DDD), Clean Architecture.
- **Frameworks & DB :** **Spring Boot**, Spring Cloud, Hibernate / JPA, RESTful APIs, **MySQL**, **Oracle**, PostgreSQL, Database Sharding & Master-Slave Replication, SQL Optimization, HikariCP Connection Pools.
- **Cloud & DevOps :** **Microsoft Azure (AZ-900 Certified)**, Docker, CI/CD (Jenkins Pipelines), CodeQL Static Analysis, SonarQube Quality Gates, Git & GitHub, JUnit, Mockito, TDD, Agile / Scrum Leadership.

PROFESSIONAL EXPERIENCE

IT Analyst (Senior Backend Engineer) @ Tata Consultancy Services

Oct 2023 – Present

- Architected and deployed scalable, event-driven microservices using Java 17, Spring Boot, and Apache Kafka, processing 10M+ daily events for healthcare claims and policy workflows with 99.99% availability.
- Implemented distributed transaction workflows across billing and policy underwriting services using the Saga Pattern (Orchestration & Choreography) and Outbox Pattern, guaranteeing eventual consistency across heterogeneous datastores without 2PC overhead.
- Engineered low-latency search and retrieval pipelines using Elasticsearch, designing custom analyzer mappings and typeahead query DSLs that reduced p99 search latency by 65% (from 450ms to <150ms).
- Refactored legacy monolithic services into modular, extensible microservices adhering to SOLID principles and GoF design patterns (Strategy, Factory, Observer, Decorator), increasing unit test coverage from 45% to 88% using JUnit and Mockito.
- Scaled database architecture by implementing MySQL/Oracle Master-Slave read replication and horizontal table partitioning, paired with multi-tier Redis distributed caching, reducing database read contention by 55%.
- Engineered system resilience and fault tolerance utilizing Resilience4j circuit breakers, token-bucket rate limiting, and graceful degradation, applying CAP and PACELC trade-offs to withstand high-volume traffic surges.
- Streamlined CI/CD deployment automation using Jenkins pipelines, enforcing automated CodeQL static analysis and SonarQube quality gates to maintain 0 critical/high vulnerabilities in production.

Application Development Senior Analyst @ Accenture

Sept 2021 – Aug 2023

- Singapore Wealth & Asset Management System: Architected backend microservices for automated portfolio rebalancing workflows, executing real-time asset allocation calculations for $500M+ in assets under management (AUM).
- Engineered secure, idempotent RESTful APIs integrating with third-party banking gateways and financial settlement networks, reducing API transaction failure rates by 40%.
- Optimized complex SQL queries, transaction boundaries, and Hibernate ORM configurations, improving end-of-day batch settlement execution time by 35%.
- **Engineering Standards & Mentorship:** Conducted Low-Level Design (LLD) reviews and mentored junior engineers on SOLID principles and clean code practices, ensuring 0 defect escapes across 4 major production releases.

PROFESSIONAL EXPERIENCE (CONTINUED)

Application Development Analyst @ Accenture
May 2020 – Aug 2021

- Australia Logistics & Order Management: Designed and developed core order processing and shipment tracking REST APIs using Java, Spring Boot, and JPA, handling 200K+ daily orders.
- Applied State and Strategy design patterns to manage multi-stage order lifecycles; optimized database indexing and HikariCP connection pools, reducing API response times by 30% during peak windows.
- **Asynchronous Event Handling:** Integrated messaging queues for asynchronous event dispatching across billing and carrier notification services, achieving 99.95% delivery success rate under surge traffic.

Associate Software Engineer @ Accenture
Feb 2018 – Apr 2020

- US Healthcare & Enterprise Case Management: Developed core Java modules, data transformation utilities, and scheduled batch jobs supporting critical public assistance workflows.
- Authored robust unit and integration test suites using JUnit and Mockito (85%+ code coverage); resolved client-reported defect tickets ahead of strict SLA deadlines.
- **Agile Delivery & RCA:** Collaborated with cross-functional teams to define user stories, perform root cause analysis (RCA), and deliver bi-weekly sprint commitments on schedule.

KEY ARCHITECTURAL PLATFORMS & DISTRIBUTED SYSTEMS

Distributed Event-Driven Insurance Platform
Java 17 • Spring Boot • Apache Kafka • Saga Pattern
High-throughput microservices handling 10M+ daily events with 99.99% availability, employing Saga choreography & Transactional Outbox for cross-service consistency.

Elasticsearch Search & Indexing Engine
Elasticsearch 7.x • Spring Data ES • Custom Analyzers
Low-latency policy and member search with custom analyzers, n-gram tokenization, and typeahead lookup, slashing p99 query latency by 65% (450ms → <150ms).

Asset Rebalancing & Investment Platform
Java • Spring Boot • Oracle DB • Idempotent REST
Real-time portfolio rebalancing engine executing automated allocations for $500M+ in assets under management (AUM) with strict idempotent guarantees and batch optimization.

Logistics Order Lifecycle Engine
Java • Spring Boot • HikariCP • State & Strategy Patterns
High-throughput order tracking and event handling pipeline managing 200K+ daily shipments with zero race conditions and 30% faster peak response times.

ARCHITECTURAL PRINCIPLES & SYSTEM DESIGN MASTERY

- **Event-Driven Microservices & Saga Pattern:** Designed asynchronous event streaming architectures with Apache Kafka; implemented Saga orchestration and choreographies with Transactional Outbox patterns to guarantee eventual consistency across distributed microservices.
- **Low-Level Design (LLD) & Clean Craftsmanship:** Strict adherence to SOLID principles and Gang of Four (GoF) design patterns (Strategy, Factory, Observer, Decorator, State); established automated CodeQL static analysis and SonarQube quality gates.
- **High-Scale Persistence & Caching Tier:** Scaled persistence architectures utilizing MySQL/Oracle read replicas, horizontal table partitioning, and multi-tier Redis caching; applied Resilience4j circuit breakers and CAP/PACELC trade-offs.
- **Distributed Consensus & Coordination:** Applied distributed systems fundamentals including Consistent Hashing, ZooKeeper coordination, distributed locking, and graceful degradation to guarantee fault tolerance under high-concurrency loads.

ENGINEERING LEADERSHIP, MENTORSHIP & GOVERNANCE

- **Technical Mentorship:** Mentored 10+ junior and mid-level engineers in Data Structures & Algorithms (DSA), Low-Level Design (LLD), SOLID principles, and clean code architecture.
- **Architecture & Design Reviews:** Spearheaded technical architecture reviews, High-Level Design (HLD) specifications, API contracts, and design doc walkthroughs for mission-critical microservices.
- **Quality & Security Champions:** Enforced test-driven development (TDD), comprehensive JUnit/Mockito test suites (88% coverage), automated SonarQube quality gates, and zero-vulnerability security pipelines.

EDUCATION

Bachelor of Engineering – Information Technology • Shri Govindram Seksaria Institute of Technology and Science (SGSITS Indore, MP) — 2013 – 2017 | **CGPA: 7.1**

Higher Secondary School Certificate (Class XII, PCM) • Vivekanand Vidya Vihar, Maral Sarovar (CBSE) — 2011 – 2012 | **87.0%**

CERTIFICATIONS & LEADERSHIP DISTINCTIONS

Microsoft Certified: Azure Fundamentals (AZ-900) • Cloud Concepts, Architecture, Governance & Compliance **[Certified]**

- **National Sports Representation:** Represented SGSITS Indore at National-Level Table Tennis Tournament (BITS Pilani Goa); Captain of state-level CBSE Football Cluster Team (2009–2011).
- **Competitive Academic Honors:** Achieved All India Rank **370 (AIR)** in the National Science Olympiad (NSO).